

Technical Design Architecture Document

1. Data Processing Architecture & Ingestion Flow

- **Automatic Format Scanning:** The system reads local folders and automatically inspects incoming file signatures via extensions (`.json`, `.txt`) to route records to respective extraction handlers without hardcoded path dependencies.
- **Extraction Processing:** Parses clean data blocks from structured applicant systems while deploying textual search extractors across messy, unstructured recruiter note pads.
- **Attribute Standardization:** Ensures irregular user entries across separate streams are corrected into a single cohesive layout pattern before warehousing.
- **Identity Merging Layer:** Utilizes unique email anchors as core matching coordinates to fuse disjointed profile items and block profile duplication.
- **Dynamic Config Projection:** Isolates structural core storage logic from visualization layers by transforming and mapping keys on-the-fly based on configurations parsed during code execution.

2. Standardized Internal Schema & Normalization Tactics

- **Deterministic Record Identification:** Implements custom namespace UUIDv5 formatting mapped tightly against the verified primary email address string to preserve data consistency.
- **Telecom Array Normalization:** Cleanses whitespace and non-numeric symbols from phone strings to consistently generate standard international `E.164` formats.
- **Integrity Validation Checks:** Employs explicit regular expression maps to test incoming strings; any malformed email structure lacking correct structural routing symbols is automatically discarded at the intake boundary.
- **Skill Dictionary Deduplication:** Trims external layout gaps, enforces total lower-case string matching, and resolves nested values into flat lists to eliminate structural identity conflicts.

3. Conflict Strategy & Reliability Calculations

- **Anchor Identification:** Relies on standardized, case-insensitive candidate email variables to serve as a uniform reference layer across systems.
- **Information Overwrites:** Enforces strict structural prioritization rules where verified fields from structured platforms automatically overwrite data fetched from manual text notes during schema collisions.
- **Integrity Telemetry Tracking:** Establishes a raw data trust configuration matrix, issuing a `1.0` maximum rating to structured APIs, while applying a `0.3` deduction penalty to free-text arrays to maintain an accurate confidence average.

4. Dynamic Layout Modification Architecture

- **Structural Decoupling:** Ensures changes to output fields do not break backend storage layers by managing all field mappings inside an independent translation matrix.
- **Schema Swapping Layer:** Reads external transformation instructions at runtime to map source keys directly into customized target visibility layouts.
- **Missing Field Action Protocols:** Operates a dedicated validation switch capable of injecting `null` strings, dropping incomplete objects entirely, or shutting down processes via precise system crash exceptions if mandatory elements vanish.

5. Production Constraints & Mitigation Plans

- **Corrupt Payloads Handling:** Embeds isolated execution exceptions around individual input items, allowing processing scripts to ignore bad blocks and log alerts safely instead of crashing entirely.
- **Text Structure Limits:** Advanced semantic entities and complex binary document tracking were explicitly removed from active scope to ensure total system speed and execution stability before submission limits